

Team TheParents Project AMI Status Report 1

Presenter: Nicolas Fernandez, NGF22

Team Members

- Stefano Buglione, SAB21S
- Lauryn Brown, LB16E
- Marie Schwarz, MCS23G
- Nicolas Fernandez, NGF22

Project Description

- Our team (called TheParents) plan to create an interactive AI model that focuses on the topics in personalized medicine.
- The AI will be called AMI (Artificial Medical Intelligence).
- Our goal is to create a functional LLM that provides source transparency.
- The AI will respond to a given prompt and provide the list of sources that were used in the response. It will provide a citation and potential link to the source.

Assignment of Responsibilities for this Week

- Stefano Buglione:
 - Draft for documentation for Functional Requirements
 - Draft for documentation for AI Avatar Requirements
- Lauryn Brown:
 - Draft for documentation for Non-Functional Requirements
- Marie Schwarz:
 - Initial draft of the SRS (Software Requirements Specification) document
- Nicolas Fernandez:
 - Creation of Use Case Diagrams
 - Identifying user roles (roles)

Current State of Project

- Up-to-date Gantt Chart:

	Project Name:	AMI (Artifical Medical Intelligence)					
	Team Members:	Stefano Buglione, Lauryn Brown, Marie Schwarz, Nicolas Fernandez					
	Start Date:	Jan-05-26					
Task #	Activity Title	Tasks	Assigned To	Start Date	Due Date	Status	Completion %
1	Team & Project Creation	Team Recruitment	-	01/05/26	01/07/26	Complete	100%
		Project Idea	-	01/05/26	01/12/26	Complete	100%
		Project Prospectus	-	01/12/26	01/23/26	Complete	100%
		Group Meetings	-	Saturday	8PM EST	In Progress	47%
2	Requirements Phase	Define system scope and constraints	-	1/12/2026	1/30/2026	Complete	100%
		Identify user roles and use cases	Nicolas Fernandez	2/1/2026	2/8/2026	Complete	100%
		Formalize functional requirements	Stefano Buglione	1/30/2026	2/5/2026	Complete	100%
		Formalize non-functional requirements(LLM, ethics, and privacy laws)	Lauryn Brown	2/4/2026	2/9/2026	Complete	100%
		Draft initial SRS (Software Requirements Specification)	Marie Schwarz	2/6/2026	2/14/2026	Complete	100%
		Define AI avatar requirements	Stefano Buglione	2/6/2026	2/14/2026	Complete	100%
3	Analysis Phase	Analyze existing AI medical systems	Nicolas Fernandez	2/15/2026	2/20/2026	In Progress	0%
		Dataset and source evaluation	Stefano Buglione	2/15/2026	2/25/2026	In Progress	0%
		Analyze bias and hallucination risks	Marie Schwarz	2/18/2026	2/28/2026	In Progress	0%
		Feasability for avatar tools analysis	Lauryn Brown	2/20/2026	3/1/2026	In Progress	0%
		Finalize system requirements	Marie Schwarz	2/20/2026	3/5/2026	In Progress	0%
4	Design Phase	System architecture design	-	3/6/2026	3/12/2026	Not Started	0%
		LLM training pipeline and deployment design	-	3/6/2026	3/18/2026	Not Started	0%
		UI/UX wireframes	Nicolas Fernandez	3/10/2026	3/20/2026	Not Started	0%
		Design AMI avatar appearance and behavior	Lauryn Brown	3/6/2026	3/22/2026	Not Started	0%
		Design source transparency and citation system	Stefano Buglione	3/15/2026	3/25/2026	Not Started	0%

Current State of Project

- Stefano's work for:
 - Functional Requirements:

A.M.I Functional Requirements (Draft)

1. **User Prompt Input:** A.M.I will be able to receive prompts through a text-based chat interface.
2. **Response Generation:** A.M.I will answer prompts with AI generated responses using a trained large language model.
3. **Medicine related responses:** A.M.I will always prioritize responding to prompts with topics related to personalized medicine, health education, and other medicine related information.
4. **No Diagnosis policy:** As an AI, A.M.I will always remind the user that in order to get a proper diagnosis it is always necessary to visit a medical professional.
5. **Source Transparency:** A.M.I will always be transparent of the sources used to generate its responses.
6. **Citation generation:** A.M.I will always cite all sources used to generate its response at the end of the generated message
7. **Web deployment:** Users will be able to access a text-based interface with A.M.I through a web application.
8. **Speech-to-text Support (Stretch Goal):** A.M.I will be able to take in speech-to-text prompts
9. **Feedback collection:** After a conversation is over, A.M.I will provide means to the user to submit feedback on its responses in order to optimize bot performance.

Current State of Project

- Stefano's work for:
 - AI Avatar Requirements:

A.M.I Avatar Requirements

1. **Avatar Display:** the application will display A.M.I as a visible virtual avatar during conversation without obstructing the user interface.
2. **Text-to-Speech Capabilities:** A.M.I will be able to use Text-to-Speech to generate both text and speech responses to the user
3. **Voice Tone and Personality:** A.M.I Will have a tone and personality befitting of its role as a A.I. medical assistant while trying to sound more like a person and less like a chatbot.
4. **Synchronization Between Speech and Text:** The speech response and the text response will be as synchronized as possible.
5. **Audio Controls:** Several audio playback controls will be available to the user to control A.M. I.'s speech such as: play/pause button, volume bar, mute button, and response replay
6. **Avatar Interactive Behavior:** A.M.I will visually give the user signals to know what state it is currently in. These states include but are not limited to: loading state, speaking state, and idle state.
7. **Avatar Appearance:** A.M.I Will have an appearance befitting of a medical assistant and a friendly design to make interaction with the user as smooth as possible.
8. **Ethical Requirements:** A.M.I will constantly remind the user that it is not a medical professional, and it does not have a license to formally diagnose any disease.

Current State of Project

- Lauryn's work for:

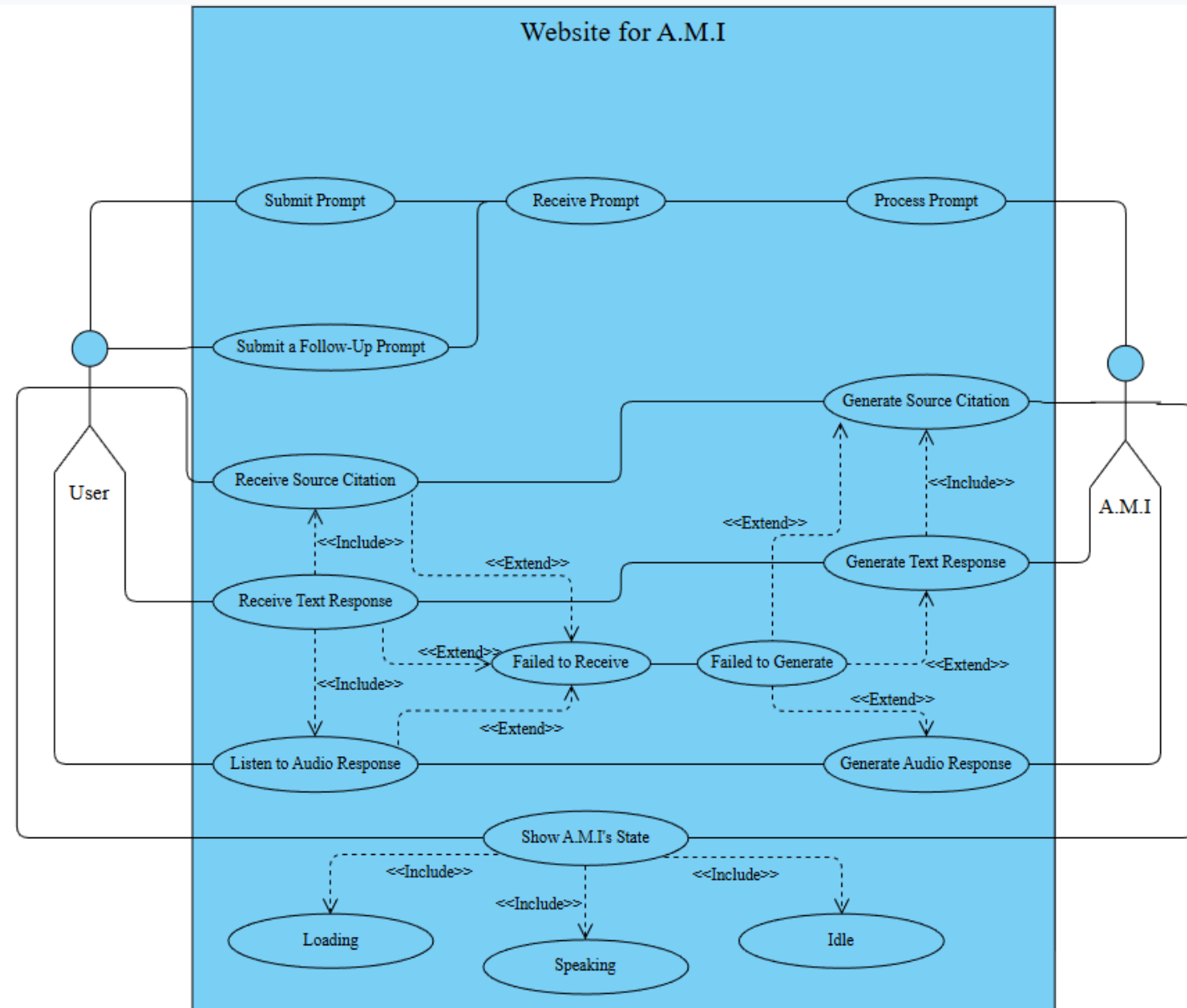
- Non-Functional Requirements:

Non-Functional Requirements

1. The system shall ensure that responses are generated using validated, peer-reviewed medical sources and shall provide visible citation references with each response.
2. The system shall not store personally identifiable information (PII) or protected health information (PHI) unless explicitly required and securely encrypted in compliance with applicable privacy standards.
3. The system shall be deployable as a web-based application compatible with modern web browsers (Chrome, Firefox, Safari, Edge).
4. The system shall achieve a minimum response accuracy rate of 90% during internal validation testing.
5. The system shall support iterative dialogue, allowing users to request clarification or refinement of responses.
6. The system shall respond to user queries within 30 seconds under normal operating conditions. (Performance)
7. The system shall maintain system uptime of at least 95% during deployment. (Reliability)

Current State of Project

- Nicolas' work for:
 - Use Case Diagrams:



Current State of Project

- Nicolas' work for:
 - A snippet of the User roles (Roles):

2. User Role:

User Permissions			
Prompt	Text	View	View_Current_Prompt
			View_Previous_Prompt
		Edit	Insert_Text
			Modify_Text
			Delete_Text
		Delete	
	Speech-To-Text	View	View_Current_Prompt
			View_Previous_Prompt
		Edit	Retake_Prompt
			Delete_Prompt
		Delete	
Response	Text	Receive	
	Speech	Receive	
	Source Citation	Receive	
Conversation	View		
Evaluations	View		
	Create		

Current State of Project

- Marie's work for:
 - A snippet of the Initial draft of the SRS:

1. Introduction

Purpose:

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Artificial Medical Intelligence (A.M.I) system. It serves as a reference for developers, testers, and stakeholders to ensure alignment throughout the software development lifecycle.

Scope:

A.M.I is a web-based AI assistant designed to provide educational information related to personalized medicine while maintaining transparency through citations from peer-reviewed sources. The system will not diagnose illnesses or prescribe treatments.

Overview:

This document outlines system functionality, interfaces, constraints, performance expectations, and ethical considerations necessary for successful deployment.

Planned Activities for the following week(s)

- Stefano Buglione:
 - Evaluation for AMI's dataset and sources
- Lauryn Brown:
 - Analysis of the feasibility for avatar tools
- Marie Schwarz:
 - Analysis of AI bias and hallucination risks
 - Finalization system requirements
- Nicolas Fernandez:
 - Analysis of existing AI medical systems

Issues not Resolved

- (1) For this week's work there have been no reported issues from the team.
- (2) (a) There is a concern on the source acquisition for AMI. In the testing phase there is a plan to place additional test data to the base dataset to have more reliable sources.
- (2) (b) A discussion was made if the base model should have exclusively journal articles or PRJAs (Peer-Reviewed Journal Articles) to make source reliability more consistent. The consensus at this current moment in time is that we first verify that the base dataset works before considering these options.

Issues not Resolved

- (3) (a) There has been a new consideration of adding another stretch goal of creating accounts for users. The original idea was for AMI to have one-off conversations with users (with no login required) to remove the need to store user data (specifically login and conversation information).
- (b) With this new stretch goal there can potentially be new functionality introduced to AMI for registered accounts such as saved conversations and (if this stretch goal is implemented to the final product) place the other stretch goal of speech-to-text prompting.
- (c) An issue is if more stretch goals are added later this may cause issues with the design for AMI. Any new ideas that pop-up in discussions will be considered as a stretch goal until further consideration.

Issues not Resolved

- (4) There has been a discussion on what the accuracy rating for prompt responses of AMI. The issue is that without testing there is no way to ensure the expected accuracy is achievable. The current accuracy rating that has been placed is 90%. Once initial testing of the AI is completed the accuracy rating will be revalidated.